

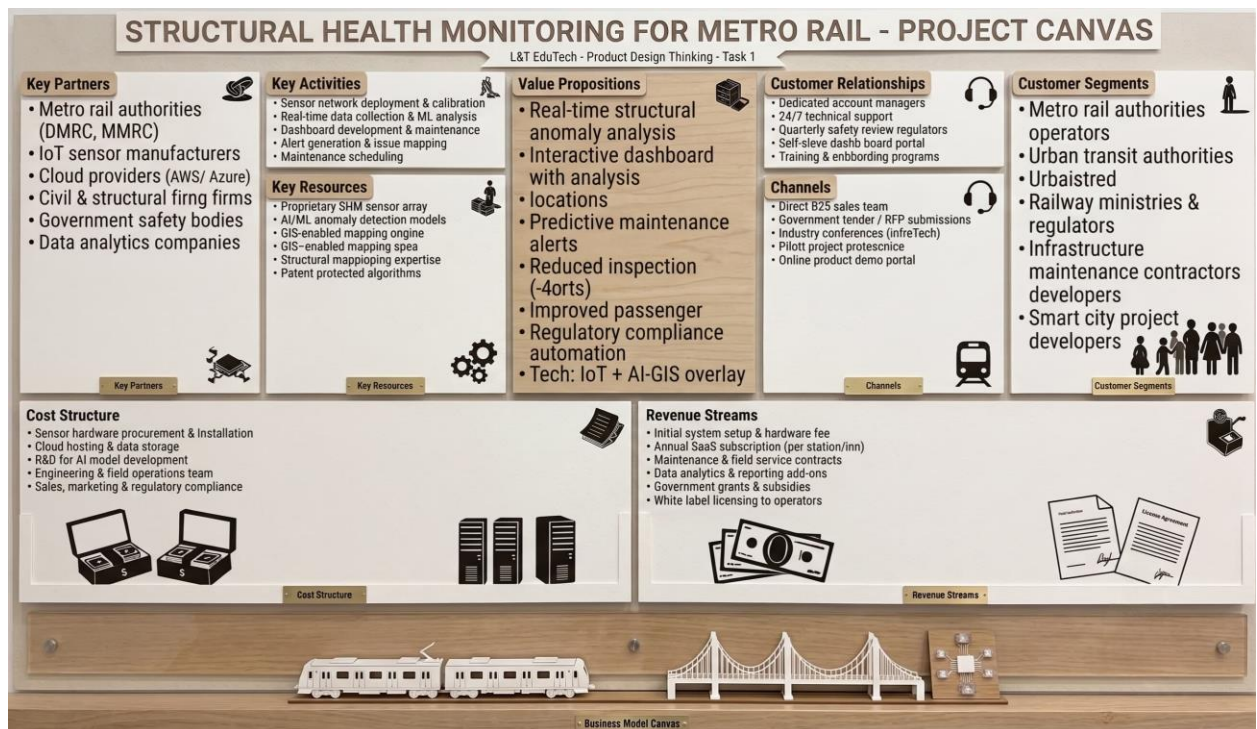
Product Design Thinking Framework – Project

Name of the student in the group: -

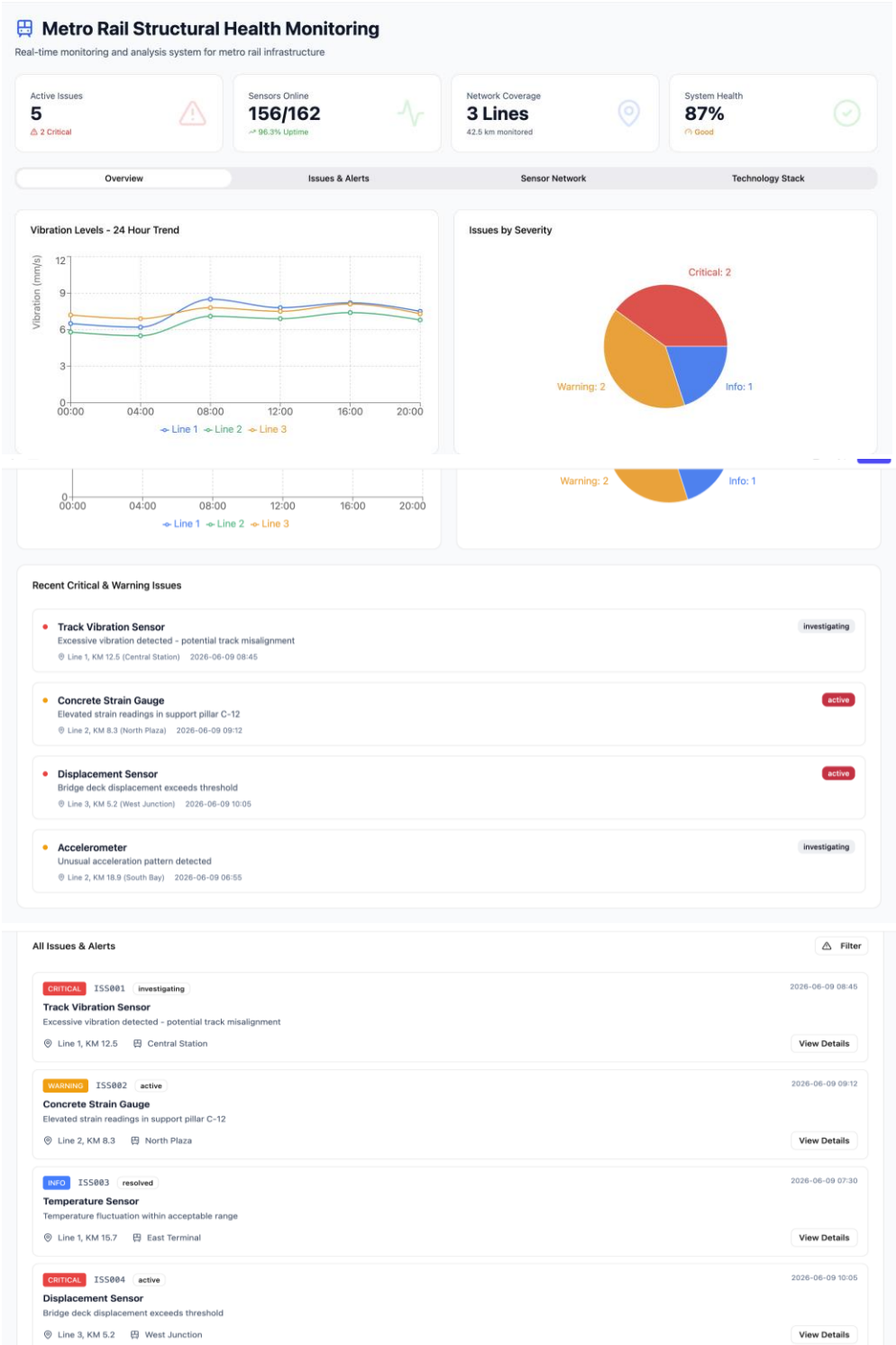
- Sachin Kumar
- Satyam Shankar Tripathi
- Abhay Kumar

A) Imagine you are a new entrepreneur aspiring to, develop a complete system overview solution for the structural health monitoring on metro rail networks, including a dashboard to display identified issues and corresponding locations, highlighting components and technologies used.

Task 01: Create a business model canvas of design thinking for the above problem



Task 02: Prepare a UI template using a suitable tool (We used Figma and figma link attached)

**Recent Critical & Warning Issues**

Track Vibration Sensor
Excessive vibration detected - potential track misalignment
Line 1, KM 12.5 (Central Station) 2026-06-09 08:45 Investigating

Concrete Strain Gauge
Elevated strain readings in support pillar C-12
Line 2, KM 8.3 (North Plaza) 2026-06-09 09:12 Active

Displacement Sensor
Bridge deck displacement exceeds threshold
Line 3, KM 5.2 (West Junction) 2026-06-09 10:05 Active

Accelerometer
Unusual acceleration pattern detected
Line 2, KM 18.9 (South Bay) 2026-06-09 06:55 Investigating

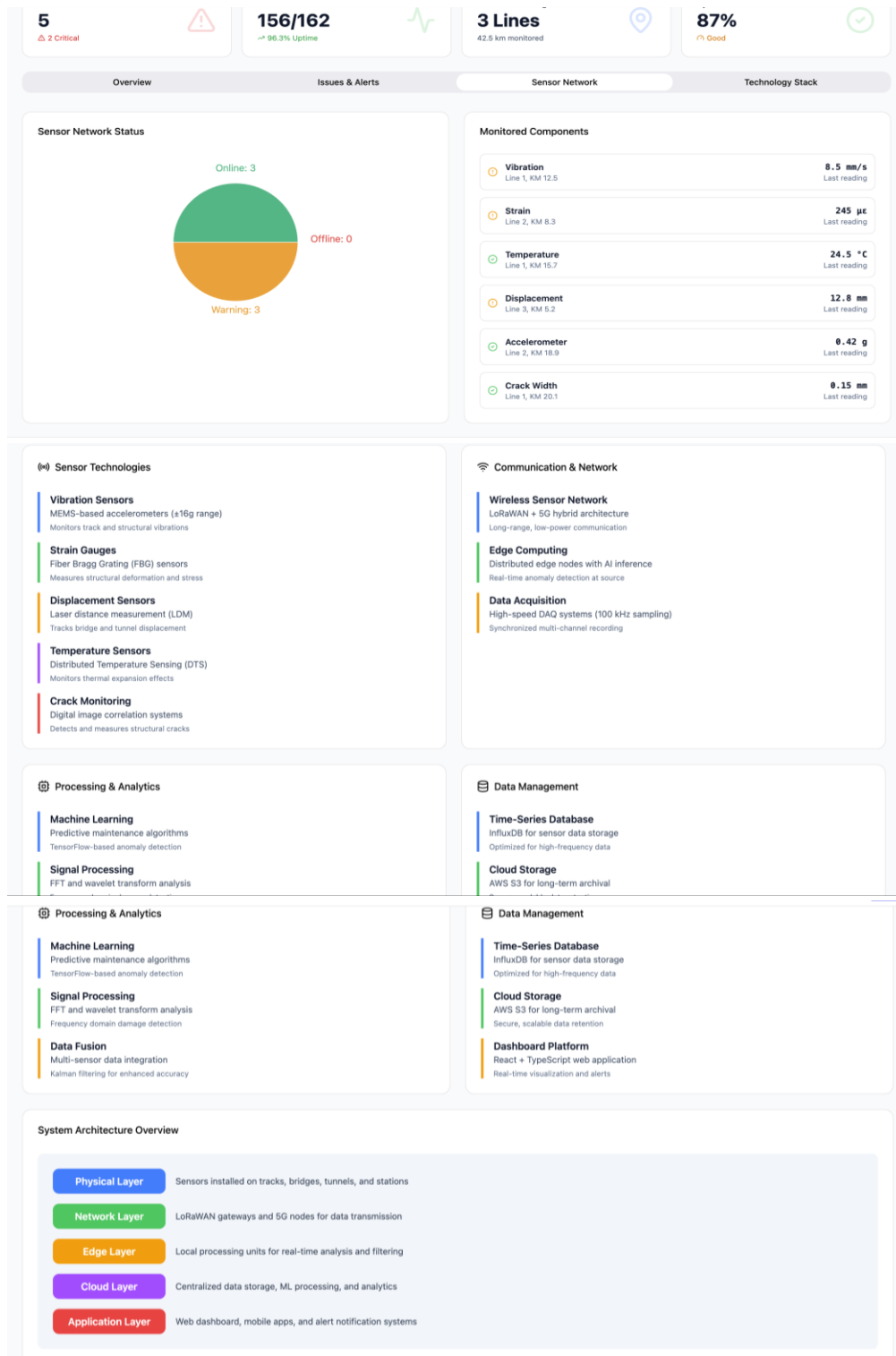
All Issues & Alerts Filter

CRITICAL ISS001 Investigating 2026-06-09 08:45
Track Vibration Sensor
Excessive vibration detected - potential track misalignment
Line 1, KM 12.5 Central Station View Details

WARNING ISS002 Active 2026-06-09 09:12
Concrete Strain Gauge
Elevated strain readings in support pillar C-12
Line 2, KM 8.3 North Plaza View Details

INFO ISS003 Resolved 2026-06-09 07:30
Temperature Sensor
Temperature fluctuation within acceptable range
Line 1, KM 15.7 East Terminal View Details

CRITICAL ISS004 Active 2026-06-09 10:05
Displacement Sensor
Bridge deck displacement exceeds threshold
Line 3, KM 5.2 West Junction View Details



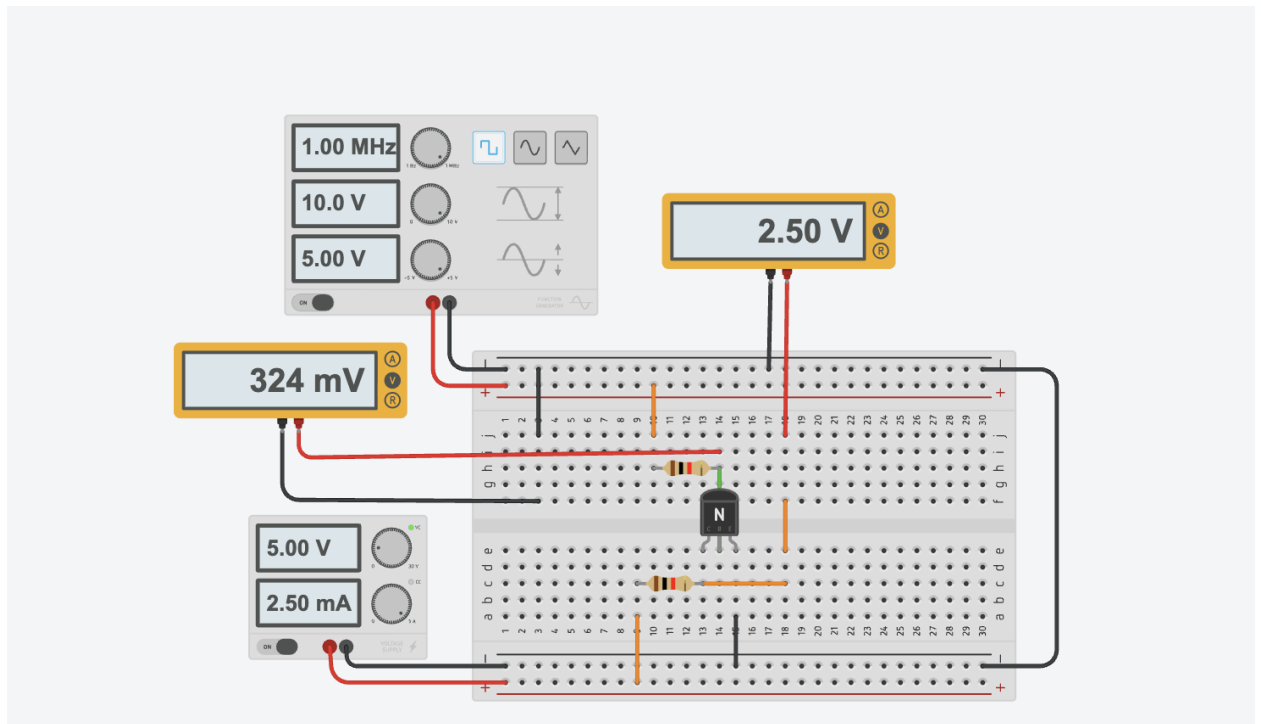
Figma Link: <https://www.figma.com/make/YIWz4FQF6XvrGCVuBepYxt/Structural-Health-Monitoring-Dashboard?t=NkxJU7ky1wV4RP6G-20&fullscreen=1>

Task 03: Create a customer journey map for the given problem

	1. AWARENESS	2. CONSIDERATION	3. DECISION	4. ONBOARDING	5. OPERATIONAL USE	6. EXPANSION
GOALS	Understand risks of manual inspection	Evaluate SHM solutions and ROI	Approve budget and vendor	Deploy sensors and dashboard	Monitor infrastructure health	Expand system across network
TOUCHPOINTS	Safety reports, Metro incidents, Industry conferences	Product demos, Case studies, Vendor meetings	Pilot project results, Procurement meetings	Installation visits, Training sessions	SHM Dashboard, Alerts, Monthly reports	ROI review meetings, Expansion proposals
ACTIONS	Reviews inspection reports, Identifies safety gaps	Compares vendors, Requests demonstrations	Conducts pilot, Selects vendor	Installs sensors, Configures dashboard	Reviews alerts, Dispatches maintenance teams	Approves rollout to additional lines
EMOTIONS	 Concerned	 Curious	 Confident	 Overwhelmed	 Satisfied	 Excited
PAIN POINTS	Manual inspections are slow, Human errors	Difficult to compare solutions, Unclear ROI	Long approval process, Budget constraints	Installation disruption, Staff training needs	Too many alerts, Data overload	Scaling costs, Integration challenges
OPPORTUNITIES	Awareness campaigns, Safety workshops	Free ROI calculator, Demo dashboard	Pilot project with low cost	Dedicated onboarding support	AI-powered predictive alerts	Multi-line discounts, Advanced analytics

B) Simulate, observe and analyze the output characteristics of a BJT common base amplifier, and explain the following

Simulation Image:



Simulation Image Link: <https://github.com/sachin91-hue/Project/blob/main/Project.mov>

i) **BJT is called a current controlled current source**

A current buffer transfers current from input to output without significant change.

In Common Base configuration:

$$I_C = \alpha I_E$$

where,

- I_C = Collector Current
- I_E = Emitter Current
- α = Current Gain ($\approx 0.95-0.99$)

From the output characteristics, for a fixed emitter current, the collector current remains almost constant and is nearly equal to αI_E .

Therefore:

- Input quantity = Current (I_E)
- Output quantity = Current (I_C)

Hence, the output current is controlled by the input current.

Therefore, BJT behaves as a Current Controlled Current Source (CCCS).

ii) CB-BJT can be used as a Current Buffer

A current buffer transfers current from input to output without significant change.

In CB configuration:

$$I_C \approx \alpha I_E$$

because α is very close to unity.

Characteristics of CB current buffer:

- Low input resistance.
- High output resistance.
- Current gain close to 1.
- Output current follows input current almost exactly.

Thus, the load receives nearly the same current as supplied at the input, independent of output voltage variations.

Therefore, the Common Base BJT acts as an efficient Current Buffer.

Result

The output characteristics of a CB amplifier show that:

- i) Collector current is controlled by emitter current, proving that a BJT acts as a Current **Controlled Current Source (CCCS)**.
- ii) Since $I_C \approx \alpha I_E$, the CB configuration transfers current with negligible loss and therefore functions as an **effective Current Buffer**.